

A *library* is a generic term for a collection of code made by a third party. A library can contain functions, classes, or other pieces of code for a developer to use. A Python library might take the form of a package, a set of packages, or even just a single module. Libraries are often specific to a particular language. The developer doesn't need to know how the library code works; they only need to know how to call or interface with the code in a library. A *standard library*, such as the Python standard library, is a code library that is assumed to be available to all implementations of a programming language.

A *framework* is a collection of code that operates with *inversion of control*; the developer creates functions that the framework will call as needed, as opposed to the developer's code calling functions in the framework. Inversion of control is often described as "don't call us, we'll call you." For example, writing code for a web app framework involves creating functions for the web pages that the framework will call when a web request comes in.

A *software development kit (SDK)* includes code libraries, documentation, and software tools to assist in creating applications for a particular operating system or platform. For example, the Android SDK and iOS SDK are used to create mobile apps for Android and iOS, respectively. The Java Development Kit (JDK) is an SDK for creating applications for the JVM.

An *engine* is a large, self-contained system that can be externally controlled by the developer's software. Developers usually call functions in an engine to perform a large, complex task. Examples of engines include game engines, physics engines, recommendation engines, database engines, chess engines, and search engines.

An *application programming interface (API)* is the public-facing interface for a library, SDK, framework, or engine. The API specifies how to call the functions or make requests of the library to access resources. The library creators will (hopefully) make documentation for the API available. Many popular social networks and websites make an HTTP API available for programs to access their services rather than a human with a web browser. Using these APIs allows you to write programs that can, for example, automatically post on Facebook or read Twitter timelines.